A GEOGRAPHIC INFORMATION SYSTEMS PROGRAM TO OPTIMIZE FEEDSTOCK LOGISTICS FOR BIOENERGY PRODUCTION FOR MOBILE PYROLYSIS UNITS

M. Ha, C. L. Munster, T. L. Provin

ABSTRACT. *A comprehensive decision-making tool was developed to optimize the use of a fleet of mobile pyrolysis units in the North Central region of the U.S. to produce bio-oil from agricultural feedstocks. The concept is to use mobile pyrolysis units to convert low-density biomass to high-density bio-oil to minimize the cost of feedstock logistics. The feedstocks evaluated in this project included corn stover, energy sorghum, and switchgrass. It was assumed that (1) energy sorghum would replace grain sorghum and (2) switchgrass would be grown on Conservation Reserve Program (CRP) land. A Geographic Information System (GIS) program was developed to optimize the movement of mobile pyrolysis units based on an analysis of transportation networks, cropping patterns, feedstock production rates, and oil refinery locations in the North Central region. The locations of existing corn, sorghum, and CRP fields in the North Central region were obtained from the USDA National Agricultural Statistics Service (NASS). Model Builder, a utility in ArcGIS, was used to automate the GIS procedures. Network Analysis, an extension of ArcGIS, was used to find the best route to move the mobile pyrolysis units to new locations and to identify the nearest refinery to receive the bio-crude oil. The feedstock input rate for each mobile pyrolysis unit was assumed to be 40 tons/day.*

Keywords. *ArcGIS Model Builder, Biochar, Bio-oil, Corn stover, Energy sorghum, Feedstock logistics, GIS, Mobile pyrolysis, Network analysis, Switchgrass.*

Increasing world gas prices and global warming are contributing to interest in renewable energy sources, including energy produced from crop residues. Pyrolysis is a thermal conversion process for converting agricultural residues and many other carbon-based materials into bio-energy. A pyrolysis system can convert diverse feedstocks to bio-crude oil, synthesis gas, and biochar. Corn stover, energy sorghum, and switchgrass are possible feedstocks for pyrolysis in the North Central region of the U.S. The goal of this study was to find optimal locations for mobile pyrolysis units and to minimize feedstock transportation costs. To accomplish this goal, a GIS program was developed to optimize feedstock logistics for the production of bio-oil using mobile pyrolysis units. The GIS analysis provided (1) optimum locations for mobile pyrolysis units based on feedstock availability, (2) feedstock hauling distances from the fields to the mobile pyrolysis units as well as biochar hauling distances back to the feedstock production fields, (3) optimum routes for moving the mobile pyrolysis units between locations, and (4) optimum routes for transporting the bio-oil to the nearest refinery. Model Builder is a utility in ArcGIS used to automate individual procedures by using model tools and scripts. Model Builder was used to automate the GIS processes and to produce a model with limited manual inputs. Network analysis was used to search for optimum routes for hauling feedstock from one mobile pyrolysis location to the next and for hauling bio-oil from the pyrolysis units to the nearest refineries.

Mobile pyrolysis units have many advantages over centralized pyrolysis plants because they can be moved to feedstock production fields to greatly simplify feedstock logistics. Biofuel production may be uneconomical due to weather, availability of feedstock, or other factors (Palma et al., 2011). In addition, some regions may be able to easily supply feedstock in the summer but not in the winter. Mobile pyrolysis units can be repositioned in the event of a crop failure and can take advantage of seasonal feedstock supplies. If one region is having difficulty supplying feedstock due to weather conditions or crop production problems (e.g., late harvest or failed crop), mobile pyrolysis units can be moved to more productive feedstock areas. A fleet of mobile pyrolysis units that can be transported quickly and easily on tractor trailers can take advantage of seasonal feedstock availability at multiple locations (Palma et al., 2011). In addition, mobile units could help improve bioenergy profitability by reducing feedstock transportation distances. According to Roberts et al. (2010), feedstock transportation distances are a significant problem for the economic viability of biochar pyrolysis systems.

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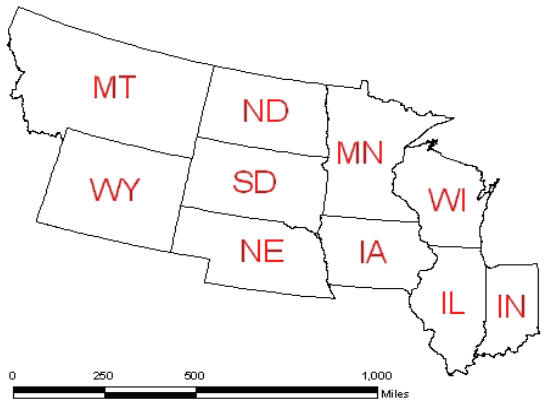


Figure 1. States in the North Central region of the U.S. included in this feedstock logistic study for mobile pyrolysis units.

This GIS project analyzed feedstock logistics for mobile pyrolysis units in the North Central region of the U.S. (fig. 1). The feedstock field locations, grain yields, and production rates were obtained from the USDA National Agricultural Statistics Service (NASS). The GIS program optimized the movement of mobile pyrolysis units in the North Central region to minimize feedstock transportation costs. The GIS program was used to assess biomass availability and feedstock transportation logistics. In previous studies, GIS-based models were used to (1) identify suitable locations for bioenergy plants based on regional demand, supply, and proximity (Kurka et al., 2012); (2) assess the technical and economic potential for biomass-based energy production (Beccali et al., 2009; Palma et al., 2011); (3) quantify the potential amount of biomass and consider the impacts of biomass removal from forest areas (Schreurs et al., 2011); and (4) combine with forest models to estimate the regional supply of harvest residues in Norway (Rørstad et al., 2010).

PYROLYSIS

Pyrolysis is the thermal degradation of biomass in the absence of oxygen. During pyrolysis, the feedstock is heated to temperatures of 400°C to 600°C and converted to bio-oil, biochar, and synthesis gas (syngas). Numerous agricultural crops can be used in pyrolysis, including corn stover, sorghum, switchgrass, rice, wheat, sugar cane, straw, and miscanthus (Mohan et al., 2006).

Pyrolysis requires drying and grinding of the feedstock to optimize heat transfer rates. After the feedstock is dried and ground, it is rapidly heated to high temperatures to produce vapors, aerosols, and biochar. The vapors and aerosols are then cooled and condensed, forming bio-oil (Bridgwater et al., 1999). The bio-oil produced by pyrolysis is typically a dark brown, organic liquid with high oxygen content (Czernik and Bridgwater, 2004). The gases that cannot be condensed are known as syngas. Syngas is mostly carbon monoxide and hydrogen and can be used to generate electric power.

The fast pyrolysis process produces mostly bio-oil, which can be upgraded to transportation fuels, along with smaller quantities of biochar and syngas (Brown et al., 2011). Fast pyrolysis uses six steps for conversion of biomass to transportation fuels: (1) pretreatment (drying biomass to 7% moisture content and grinding it to a final particle size of 3 mm), (2) pyrolysis (fluidized bed reactor operation at 450°C and atmospheric pressure in an oxygen-free environment), (3) solids removal (biochar), (4) oil recovery (using an indirect heat exchanger and electrostatic precipitator to collect condensable vapors), (5) heat generation, and (6) hydroprocessing (Brown et al., 2011).

Biochar is a charcoal-like substance that is a by-product of the pyrolysis process. After bio-oil, biochar is the second highest yielding product for both slow and fast pyrolysis systems, typically in the range of 15% to 40% of the biomass feedstock on a weight basis (Brown et al., 2011). Biochar additions to soils may mitigate some of the negative effects of removing crop residues from production fields and enhance the soil quality. Applying biochar to the soil replaces carbon and many of the plant nutrients that are removed from the soil with the biomass (Mullen et al., 2010).

MOBILE PYROLYSIS

Mobile pyrolysis units have many advantages over large centralized pyrolysis plants. Mobile pyrolysis units can be deployed directly to the feedstock production fields, thereby minimizing feedstock hauling distances. Agricultural feedstocks commonly have low bulk densities that range from 40 to 200 kg/m^3 (Adapa et al., 2011) and high water contents of 10% to 85% (Propheter et al., 2010). A mobile pyrolysis unit requires space only for temporary storage of the feedstock, bio-oil, and biochar. The mobile pyrolysis unit converts the energy in the feedstock into high-density bio-oil. Transporting the bio-oil over long distances to an oil refinery is more cost-effective than transporting the low-density feedstock over long distances to a centralized pyrolysis plant. In addition, the syngas produced during pyrolysis can be used to dry the feedstock and power the pyrolysis equipment. The biochar byproduct is also a low-density material that cannot be transported cost-effectively over long distances. Therefore, mobile pyrolysis units also facilitate the land application of biochar to the feedstock production fields.

The mobile pyrolysis unit used in this study was assumed to be a fluidized bed system (30.48 cm diameter) operating at a feedstock rate of 40 tons per day with a feedstock moisture content of 10% or less. The system produces bio-oil at a rate of 7,570 L per day and biochar at a rate of 10 tons per day (S. Capareda, 2009, personal communication). Fast pyrolysis systems generally produce 10% to 30% biochar on a feedstock weight basis (Brewer et al., 2012). Fast pyrolysis thermally transforms the biomass into bio-oil (~60% of mass), syngas (~20% of mass), and biochar (~20% of mass) (Laird, 2008). Brown et al. (2011) found that a fast pyrolysis system produced 567 tons/day of biochar and 1,168 tons/day of bio-oil when the biomass input rate was 2,000 tons per day.

FEEDSTOCKS

The feedstocks used in this study were corn stover, energy sorghum, and switchgrass. Corn stover was chosen because of its high production rates throughout the North

Central region. Energy sorghum was selected for analysis because it was designed specifically for high biomass production for biofuels. It was assumed that energy sorghum would be planted in fields where grain sorghum is now grown. Switchgrass was evaluated since it can be grown on the poor soils often found in Conservation Reserve Program (CRP) fields and would not displace agricultural crops. Corn stover biomass was calculated using equation 1:

$$\text{Corn stover biomass (kg)} = \text{ten-year average corn yield (bu/m}^2\text{)} \times 25.4\ \text{(kg/bu)} \times 0.25 \times \text{area (m}^2\text{)} \quad (1)$$

The ratio of corn grain (kg) to corn stover (kg) was assumed to be 1:1, and one bushel of corn was assumed to be 25.4 kg (56 lbs) (USDA, 2010b). However, only 25% of the corn stover was made available as a feedstock for pyrolysis, leaving 75% of the corn stover in the field for erosion prevention and soil improvement. Energy sorghum was assumed to yield approximately 5.6 Mg/ha, with 100% available for harvest (D. Baltensperger, 2012, personal communication). For switchgrass, mean annual yields were 2.24 Mg/ha for western Nebraska (D. Baltensperger, 2012, personal communication).

Biochar Utilization

One possible use of the biochar is to return it to the feedstock production fields as a soil amendment. Biochar contains nutrients that potentially could be utilized for feedstock production. Crop residues contain substantial amounts of plant nutrients such as C, N, K, P, Ca, and Mg, and some of these nutrients end up in the biochar (Laird, 2008). However, crop growth may be reduced when there are very high applications of biochar (Lehmann et al., 2006). Application of biochar to soils is hypothesized to increase water-holding capacity, increase saturated hydraulic conductivity, build soil organic matter, improve nutrient cycling, decrease bulk density, and reduce the transport of pesticides and nutrients to surface water and groundwater (Laird, 2008). A sustainable feedstock production system for bioenergy production will require that the biochar is land-applied to the production fields.

Literature Review

GIS analyses have been used in many previous bioenergy studies. Singh et al. (2008) studied the potential of using agricultural biomass for power production in the Indian state of Punjab. Their spatial study concluded that the costs of collection and transportation of feedstocks were important for sustainable biomass energy conversion facilities. Biomass feedstock collection and transportation costs could be reduced by locating the biomass power plants near biomass collection centers. In addition, spatial information technologies have led to more scientific planning in bioenergy power plant construction. Shi et al. (2008) presented a case study using remote sensing and GIS to evaluate the feasibility of setting up biomass power plants and optimizing the plant location in Guangdong, China. Vegetation type, ecological sustainability, economic conditions, and harvest costs were considered in the study, which used MODIS/Terra remote sensing data. A GIS program was used to assess feedstock transportation distances along roads.

In another GIS study, historic crop yields were estimated for ten sites in the Peace River region of Alberta, Canada. This study determined inter-year variability of biomass availability by quantifying the feedstock supply risk to ensure future crop productivity (Stephen et al., 2010). Biomass availability was assessed as a function of grain yield, biomass-to-grain ratio, cropping frequency, and residue retention rate. Stephen et al. (2010) found that the availability of feedstock quantities was highly dependent on the biomass-to-grain ratio and developed a methodology for determining the range of biomass availability. A spatially explicit bioenergy siting model (BSM) of the bioenergy production system in California was developed by Tittmann et al. (2010). This work expanded on previous bioenergy siting work by optimizing the system using spatially explicit feedstock supply curves, multiple potential conversion technologies, and geographically determined bioenergy demands. The study combined a spatial model with economic aspects of bioenergy production systems and used a transportation network GIS analysis with a mixed-integer linear programming (MIP) optimization model.

Beccali et al. (2009) used a GIS-based method to evaluate the energy potential of biomass from the agricultural and forestry sectors in Sicily. The GIS program considered all the transportation components for bioenergy production, including proximity to streets, morphology and elevation of terrain, density of facilities, and other factors. Ma et al. (2005) produced a siting suitability map identifying the most suitable locations for distributed bio-energy systems using dairy manure in Tompkins County, New York. The GIS model integrated both spatial and non-spatial data. The GIS model found the availability of renewable energy resources based on the distribution of dairy farms. Perpiñá et al. (2009) applied a grid concept to determine the optimal use of agricultural and forest residues. Potential sites for obtaining biomass energy and optimal locations for bioenergy plants were mapped to calculate the amount of residue within a 1 km^2 rectangular area in Valencia, Spain, using ArcGIS. Haddad and Anderson (2008) used GIS technology to identify potential locations for the collection and storage of corn stover in the Midwestern region of the U.S. Their study included the following steps: (1) determining site selection criteria, (2) identifying the study area and service area based on transportation network analysis, (3) reclassification of spatial layers, (4) overlaying the reclassified spatial layers with equal weights to produce two main models, and (5) overlaying the main models using different weights.

The centroid point concept was used to determine the routes from cotton fields to the nearest gin in Lubbock County, Texas (Simpson et al., 2007). The transport routine used local, county, and state roads. Frombo et al. (2009) used a GIS-based Environmental Decision Support System (EDSS) to define planning and management strategies for the optimal logistics for energy production from wood residues. The EDSS was divided into three modules: GIS, data

management, and optimization (for decision problems such as strategic planning, tactical planning, and operational management). A two-stage methodology was used to identify the optimum location for a facility using woody biomass in the village of L'Anse in Baraga County in the Upper Peninsula of Michigan (Zhang et al., 2011). Stage I used a GIS approach to identify feasible biofuel facility locations based on railroad and road transportation networks; county, city, and village boundaries; population densities; and pulpwood production locations. Stage II selected the preferred location using a total transportation cost model.

METHODS

A GIS program was developed to determine the optimum locations for mobile pyrolysis units in the North Central region based on feedstock availability. Move times for the mobile pyrolysis units were assumed to be 1, 2, 4, 6, 8, 10, and 12 month intervals. Feedstock availability was systematically calculated throughout the region using a square overlay grid. Grid sizes varied according to move times and feedstock production rates. ArcGIS was used to combine crop yields, field locations, and grid layers to define the biomass available in each grid cell based on pyrolysis unit move times. Corn, sorghum, and switchgrass (on CRP land) field locations were obtained from 2008 NASS cropland data layer (CDL) data. The data were provided in raster files, which were converted to shape files. Average corn stover yields were based on the ten-year average (1999 to 2008) from NASS for each county. Energy sorghum and switchgrass were assumed to yield 5.6 and 2.24 Mg/ha, respectively, for western Nebraska, with 100% available for harvest (D. Baltensperger, 2012, personal communication).

It was assumed that the mobile pyrolysis units were located in the center of a grid cell and would be able to process all of the feedstock within the cell before moving to another location. Therefore, the average distance required to transport feedstock to the mobile pyrolysis unit and to transport biochar back to the feedstock production fields was one-half the grid cell size. ArcGIS Model Builder was used to automate these GIS procedures.

The user selected the optimum mobile pyrolysis unit locations based on the GIS results for feedstock availability in each grid cell. Once the locations of the mobile pyrolysis units were identified, the shortest routes between optimum locations were determined using county-level roads and streets obtained from the U.S. Census Bureau. These line and point files were used as input for the Network Analysis extension in ArcGIS. Network Analysis was used to calculate distances from one mobile pyrolysis location to the next. Distances to the nearest oil refineries were also calculated. Model Builder was then used to automate these GIS procedures. The final GIS table included feedstock hauling distances, distances from one location to the next, and distances to the nearest oil refineries.

INPUT LAYERS

The locations where corn and sorghum fields were planted in the North Central region in 2008 were obtained from the 2008 CDL database (USDA, 2010a). NASS provides the annual CDL data with crop-specific digital data layers in GIS raster format (USDA, 2010a). The CDL program uses imagery from the Resourcesat-1 AWiFS and the Landsat 5 satellites, producing digital categorized georeferenced images exported in GeoTiff format for use in the ArcGIS interface. The Institute of Natural Resources Sustainability at the University of Illinois at Urbana-Champaign performed a CDL accuracy assessment for remote sensing applications. The analysis found strong evidence of land cover classification success, with 97.6% accuracy, a corresponding omission error of only 2.4%, and an overall kappa coefficient of 0.95 (Luman and Tweddale, 2008). The CDL raster data for corn and sorghum fields planted in the North Central region were converted to polygon shape files to identify feedstock production locations.

The transportation networks in the North Central region were used to optimize feedstock logistics. A network database that contains all the interstate highways, major roads, and local roads in the North Central region was developed. The road network data were available from the U.S. Census Bureau at ftp://ftp2.census.gov/geo/tiger (U.S. Census Bureau, 2009a), which provides ArcGIS formatted roads and streets on a county basis. A shape file with county-level roads and streets was obtained from the U.S. Census Bureau at ftp://ftp2.census.gov/geo/tiger/TIGER2007FE (U.S. Census Bureau, 2009b). Major highways and local roads were merged to create the transportation system using the append function in ArcGIS. The appended road lines were used as input for the Network Analysis extension in ArcGIS to create the road networks.

GRID CELLS

A GIS program was developed to create a square grid that overlaid the entire North Central region. The feedstock availability was calculated for each grid cell using the following methodology. First, a square grid was developed, with the size of the grid cells based on pyrolysis unit move times and feedstock availability. For example, a move time of six months requires more feedstock, and therefore a larger cell, than a two-month move time. The area of the feedstock fields in each grid cell was then determined using polygon shape files of planted fields in 2008 (USDA, 2010a). The grid and the feedstock shape files were merged, and shape files that overlapped multiple grid cells were subdivided into shape files that did not overlap multiple cells. The amount of corn stover available in each grid cell was calculated by multiplying the feedstock area by the ten-year (1999 to 2008) average production rate (on a county basis) from Quick Stats 1.0 (USDA, 2010b). For energy sorghum, the feedstock production rate was 13,388 lbs/acre with the assumption of 100% harvest. Mean annual switchgrass yields of 6.3 Mg/ha for Nebraska, South Dakota, and North Dakota were reported by Kiniry et al. (2008). Figure 2 demonstrates the GIS method for creating and merging the grid with corn field locations. The grid cell was laid over the feedstock shape files, and the feedstock production area was calculated in each grid cell.



(a) Grid (b) Feedstock locations (c) Merged grid and feedstock locations

Figure 2. GIS method used to calculate feedstock availability for mobile pyrolysis units: (a) a square grid was developed, (b) planted feedstock fields as polygon shape files (raster files converted to vector files) from NASS were used, and (c) the grid and feedstock shape files were merged. Shape files that overlapped multiple grid cells were subdivided into shape files that did not overlap multiple cells.

CORN STOVER

Crop yields on a county basis were obtained from NASS (USDA, 2010b) and used in this study. For example, the ten-year average (1999 to 2008) corn grain yield in LaSalle County, Illinois, was 405 bu/ha (164 bu/acre), and it was assumed that one bushel of corn grain weighed 25.4 kg (56 lbs) (USDA, 2010b). Therefore, for LaSalle County, an average grain production of 10,292 kg/ha (9,182 lbs/acre) was used. It was also assumed that 1 kg of corn grain also produced 1 kg of corn stover biomass (Pordesimo et al., 2004). However, only 25% of the corn stover was made available for the pyrolysis units, leaving 75% on the land to protect against erosion and to improve the soil. Therefore, an average of 2,572 kg/ha (2,295 lbs/acre) of corn stover was assumed to be available as feedstock for the mobile pyrolysis units in LaSalle County. Since a mobile pyrolysis unit utilizes feedstock at a rate of 36,287 kg/day (80,000 lbs/day) at 10% moisture content, this is equivalent to a utilization rate of 14.1 ha/day (34.85 acres/day = 80,000 lbs/day ÷ 2,295 lbs/acre). On an annual basis, the area of corn stover required to supply a mobile pyrolysis unit for one year in LaSalle County was 5,148 ha (12,721 acres = 34.85 acres/day × 365 days). Several move time scenarios (1, 2, 4, 6, 8, 10, and 12 months) were applied to the mobile pyrolysis units. Average crop yields and grain production rates as well as the corn stover acreage required to supply mobile pyrolysis units for various move times were calculated in the GIS analysis for the entire North Central region.

ENERGY SORGHUM

It was assumed that a high-biomass cultivar of sorghum (energy sorghum) would be grown in fields where grain sorghum is currently planted in the North Central region. The basic assumption for energy sorghum is that 100% would be harvested each year. The biomass production rate was assumed to be 5.6 Mg/ha (2.5 tons/acres) (D. Baltensperger, 2012, personal communication). Therefore, based on a production rate of 5.6 Mg/ha, the mobile pyrolysis unit utilized 6.47 ha/day (16.0 acres/day = 80,000 lbs/day ÷ 4,996 lbs/acre) of sorghum feedstock, and the area of energy sorghum required to supply a mobile pyrolysis unit for one year was 2,365 ha (5,844 acres = 16.0 acres/day × 365 days). The area of energy sorghum required to supply a mobile pyrolysis unit for move times of 1, 2, 4, 6, 8, and 10 months was calculated using 6.47 ha/day (16.0 acres/day).

SWITCHGRASS

Switchgrass was one of the possible feedstocks considered in this study, but very little switchgrass is currently grown in the U.S. However, some of the CRP lands in the North Central region may be a good source of switchgrass in the future. The CRP protects millions of acres of American topsoil from erosion and is designed to safeguard the nations' natural resources (USDA, 2011). In general, perennial dry land forages in the region were less than 2.24 Mg/ha (1 ton/acre), and this value was used for switchgrass yields (D. Baltensperger, 2012, personal communication). At a feedstock input rate of 40 tons/day (80,000 lbs/day), a mobile pyrolysis unit utilized 16.2 ha/day (40.0 acres/day). The area of switchgrass required to supply a mobile pyrolysis unit for one year was 5,913 ha (14,611 acres = 40.0 acres/day × 365 days).

MODEL BUILDER

ArcGIS Model Builder was used to automate the GIS procedures to analyze the logistics for mobile pyrolysis units. Model Builder automates individual procedures by using model tools and scripts. Figure 3 is a flowchart of the Model Builder procedure used to determine feedstock availability for mobile pyrolysis units. This procedure includes: (1) merging the grid files and crop field shape files, (2) dissolving the merged files for each grid cell, and (3) calculating the planted area and biomass production within each grid cell. Figure 4 is a flowchart of the Model Builder procedure used to generate the route and calculate the distance between two mobile pyrolysis units and the distance to the nearest oil refinery. This procedure includes: (1) using the planted areas within each grid cell and the zonal geometry function to find the centroid locations, (2) converting the centroid locations from raster format to point files, (3) importing the road network data to Network Analysis, (4) transferring the centroid points to Network Analysis, and (5) calculating the shortest route.

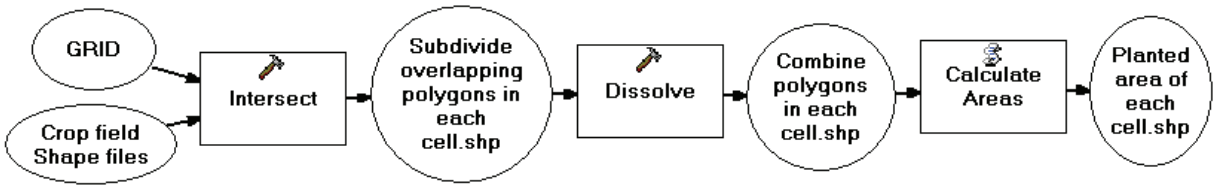


Figure 3. Flowchart of Model Builder procedure used to determine feedstock availability for mobile pyrolysis units in the North Central region.

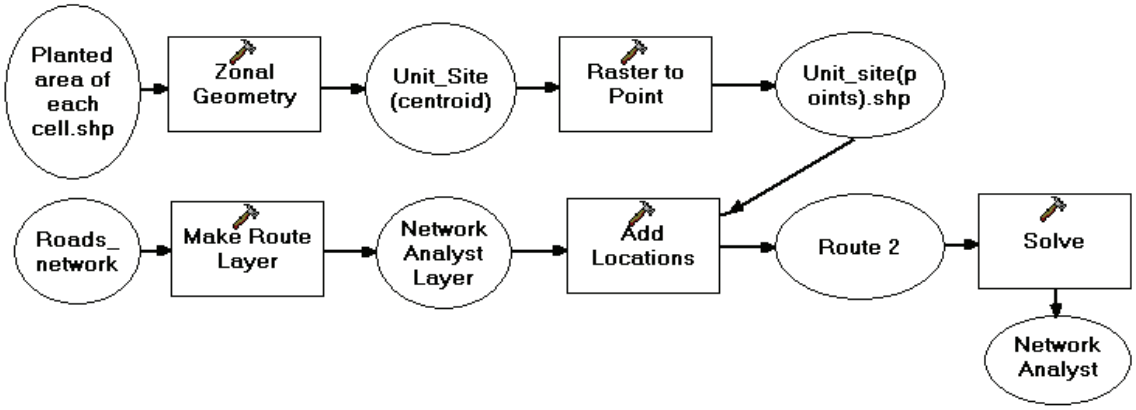


Figure 4. Flowchart of Model Builder procedure used to route the mobile pyrolysis units from one location to the next and determine the distances from the mobile pyrolysis location to the nearest oil refinery.

It was assumed that the feedstock would be collected at the centroid point of each grid cell. The routing distance calculations started from the centroid point to the next mobile pyrolysis location or the nearest oil refinery. Since the centroid point may be in the middle of a field, there may be a gap between the centroid point and the transportation starting point on the road network. This gap from the centroid point where feedstock was collected to the closest point on the road network varied from 24 to 661 m. These small distances were neglected in the transportation distance calculations.

Network Analysis

Flexibility and simplified feedstock logistics are major strengths of using the mobile pyrolysis concept to produce bio-oil. A basic assumption was that the mobile pyrolysis units were located in the center of each grid cell. Therefore, the average distance for feedstock to move to a mobile pyrolysis unit was assumed to be half the grid cell size. It was also assumed that the average distance required to transport biochar back to the feedstock production fields was half the grid cell size. The Network Analysis extension in ArcGIS was used to calculate: (1) the distances from one mobile unit location to the next, and (2) the distances to the nearest oil refinery. Model Builder was used to automate these processes. Figure 4 shows a flowchart of the Model Builder procedure used to generate the route from one mobile pyrolysis unit to the next based on the road network and the planted area and yields of each cell. The feedstock areas were polygon files that were used to determine the centroid point in each grid cell using the zonal geometry function in Spatial Analysis. The gap between the centroid point, where feedstock was collected, and the closest point on the road network was ignored because that distance was less than 1 km.

The raster center area was then converted to a point. These points were input to Network Analysis to calculate the routes using the shortest distance option. The routing layer (lines) was developed to find the optimal route between pyrolysis locations (points) or to the nearest oil refineries (points). The goal of optimizing the routes using Network Analysis was to minimize transportation costs based on the distances between two locations. In addition, facilities and labor charges were more sensitive than gas prices. For this project, the roadway system included one type of road network using line shape files in ArcGIS. Estimates of total costs were based on the following factors from the economic analysis (Palma et al., 2011): labor cost of truck driver (per day), feedstock loading and uploading costs, insurance costs (per month), diesel fuel cost, fuel efficiency, biomass transport distance, number of trips (40 tons/haul), and number of days.

Software and Hardware

ESRI ArcMap 9.3 version and ArcGIS Desktop 9.3 Service Pack 1 were used for the GIS analyses. This software was run on a Dell Precision T7500 workstation with a Quad Core Intel Xeon Processor E5504 (2.0 GHz, 4 M L3 cache, 4.8 GT/s). The memory was 4 GB, and a 1.0 GB graphics card was used to support high-resolution images in ArcGIS.

Results

The GIS program utilizes a combination of automatic and manual procedures. Model Builder run times varied based on the pyrolysis move times (1 to 12 months), which in turn affected grid sizes. For the entire North Central region, the run times for creating grids varied from 30 min (for 10,000 m grid size) to three days (for 1,000 m grid

Table 1. Square grid sizes based on mobile pyrolysis unit move times (1 to 12 months) for corn stover, energy sorghum, and switchgrass feedstocks that were used in the GIS analyses.

Move Time (months)	Grid Size (m)		
	Corn Stover	Energy Sorghum	Switchgrass
1	2,400	5,000	3,300
2	3,500	8,100	5,000
4	4,500	12,000	7,000
6	6,200	15,000	8,500
8	7,300	19,000	9,800
10	8,100	22,000	11,000
12	9,600	23,000	12,000

size). The total number of mobile pyrolysis units for each state can be calculated by feedstock availability and the feedstock input rate for mobile pyrolysis units. For example, the total corn stover feedstock area in Illinois for 2008 was approximately 43,949 million m^2, and the feedstock input rate for each mobile pyrolysis unit was assumed to be 40 tons/day (S. Capareda, 2009, personal communication). If all the corn stover feedstock were processed by mobile pyrolysis units in Illinois, a total of 853 mobile pyrolysis units would be needed. Details about the routing of mobile pyrolysis and the total number of mobile pyrolysis units needed will be presented in a separate report.

Determining the Grid Sizes

Square grids, based on the area required to supply feedstock to a mobile pyrolysis unit for 1, 2, 4, 6, 8, 10, and 12 month move times, were developed for the North Central region (table 1). These grids were placed over the areas with corn and sorghum production fields. The grid size for corn stover ranged from 2,400 to 9,600 m, the grid size for energy sorghum ranged from 5,000 to 23,000 m, and the grid size for switchgrass ranged from 3,300 to 12,000 m. The grid size was based on the area required to supply feedstock to a mobile pyrolysis unit based on move times of 1 to 12 months.

Routing the Mobile Pyrolysis Units

Next, Model Builder calculated the shortest distance routes using Network Analysis for (1) moving the mobile pyrolysis unit from one location to the next, and (2) transporting the bio-oil to the nearest oil refinery. Figure 5 shows the nodes (points) and links (lines) in Network Analysis, which were obtained from road and street line files in the U.S. Census Bureau ftp server. The two main object types in Network Analysis are nodes and links. Links show the relationships between nodes. A link always connects two junctions. In Network Analysis, run times for selecting locations and calculating transportation routes were relatively short (less than 5 min). Figure 6 shows the optimum routes calculated by Network Analysis to move a mobile pyrolysis unit between corn stover feedstock locations 10, 11, and 12 for a one-month move time in Carroll County, Illinois. The routing method was based on finding the shortest distances between pyrolysis locations.

Transporting Bio-oil to the Nearest Refinery

The optimum route for transporting bio-oil from each mobile pyrolysis unit to the nearest oil refinery was determined using Network Analysis. Figure 7 shows the routing

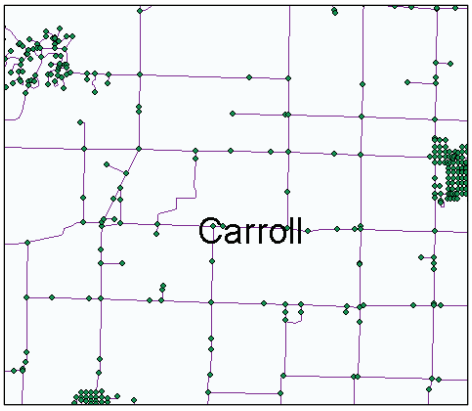


Figure 5. Network analysis using road network data to create links and nodes in Carroll County, Illinois. Links and nodes were used to calculate the shortest routes between mobile pyrolysis units and the shortest route from pyrolysis units to the nearest oil refinery.

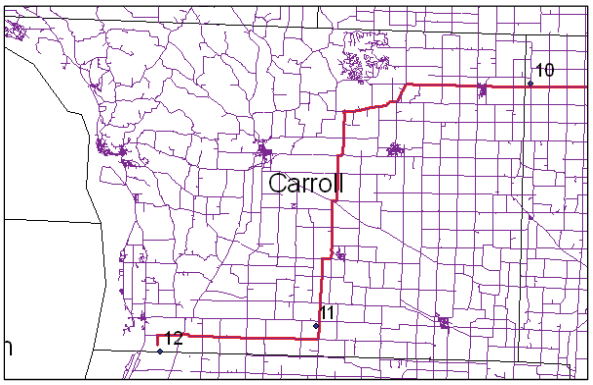


Figure 6. Example of Network Analysis output for routing a mobile pyrolysis unit between locations 10, 11, and 12 for corn stover feedstock with a one-month move time in Carroll County, Illinois.

of bio-oil from pyrolysis locations in Nebraska to the nearest oil refinery. These pyrolysis locations are the optimum locations for energy sorghum feedstock in the North Central region. The shortest routing distance option was selected.

Conclusions

A GIS program was developed to optimize the use of mobile pyrolysis units in the North Central U.S. to produce bio-oil from agricultural feedstocks. A semi-automated GIS program was developed to determine the optimum locations, based on feedstock availability, for mobile pyrolysis units to produce bioenergy in the North Central region. A methodology to determine the shortest distances between mobile pyrolysis locations and the optimum route from the pyrolysis location to the nearest oil refinery was also determined. The feedstocks evaluated were corn stover, energy sorghum, and switchgrass.

The final output table from the GIS analysis included the grid size, the feedstock and biochar transport distances between the feedstock production fields and the mobile pyrolysis units, the shortest distance from one mobile pyrolysis location to the next, and the distance from each mobile pyrolysis location to the nearest oil refinery. This GIS program is flexible and can be applied to other regions and to

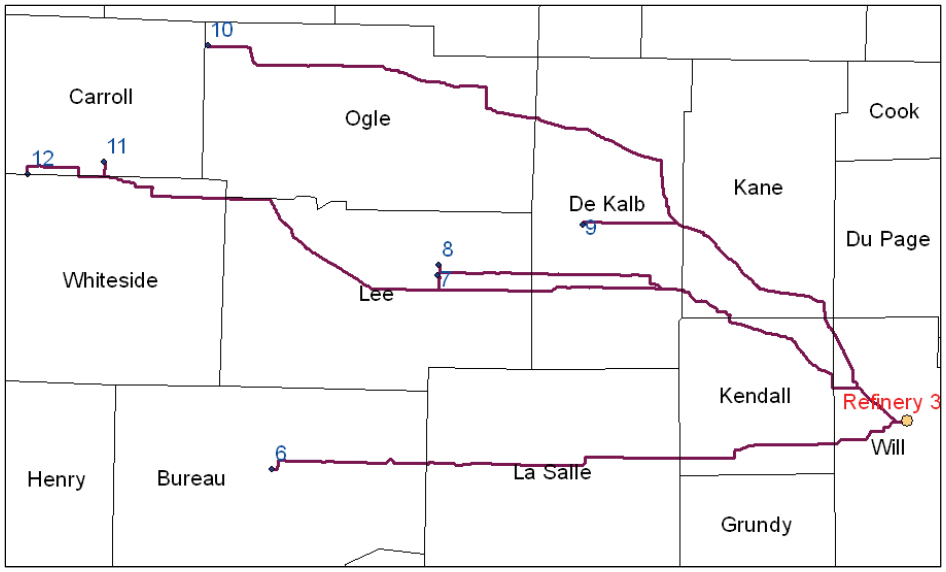


Figure 7. Example of Network Analysis results for routing bio-oil from mobile pyrolysis locations (6 to 12) to the nearest oil refinery (refinery 3) for energy sorghum feedstock in Nebraska.

other agricultural feedstocks. In future work, the movement of mobile pyrolysis units will be integrated with weather prediction models. This will allow weather information and crop production forecasts to be factored into the management process to identify optimum locations for the mobile pyrolysis units. Therefore, the GIS model will need to be interfaced with crop production predictions and meteorological forecasts on a real-time basis to schedule the movement of the mobile pyrolysis units. In addition, the GIS model will be linked to bioenergy economic models as well.

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References

Adapa, P., Tabil, L., & Schoenau, G. (2011). Grinding performance and physical properties of non-treated and steam-exploded barley, canola, oat and wheat straw. *Biomass & Bioenergy, 35*(1), 549-561. http://dx.doi.org/10.1016/j.biombioe.2010.10.004.

Beccali, M., Columba, P., D'Alberti, V., & Franzitta, V. (2009). Assessment of bioenergy potential in Sicily: A GIS-based support methodology. *Biomass & Bioenergy, 33*(1), 79-87. http://dx.doi.org/10.1016/j.biombioe.2008.04.019.

Brewer, C. E., Hu, Y. Y., Schmidt-Rohr, K., Loynachan, T. E., Laird, D. A., & Brown, R. C. (2012). Extent of pyrolysis impacts on fast pyrolysis biochar properties. *J. Environ. Quality, 41*(4), 1115-1122. http://dx.doi.org/10.2134/jeq2011.0118.

Bridgwater, A., Meier, D., & Radlein, D. (1999). An overview of fast pyrolysis of biomass. *Organic Geochem., 30*(12), 1479-1493. http://dx.doi.org/10.1016/S0146-6380(99)00120-5.

Brown, T. R., Wright, M. M., & Brown, R. C. (2011). Estimating profitability of two biochar production scenarios: Slow pyrolysis vs. fast pyrolysis. *Biofuels Bioprod. Biorefining, 5*(1), 54-68. http://dx.doi.org/10.1002/bbb.254.

Czernik, S., & Bridgwater, A. V. (2004). Overview of applications of biomass fast pyrolysis oil. *Energy Fuels, 18*(2), 590-598. http://dx.doi.org/10.1021/ef034067u.

Frombo, F., Mindardi, R., Robba, M., Rosso, F., & Sacile, R. (2009). Planning woody biomass logistics for energy production: A strategic decision model. *Biomass Bioenergy, 33*(3), 372-383. http://dx.doi.org/10.1016/j.biombioe.2008.09.008.

Haddad, M. A., & Anderson, P. F. (2008). A GIS methodology to identify potential corn stover collection locations. *Biomass Bioenergy, 32*(12), 1097-1108. http://dx.doi.org/10.1016/j.biombioe.2008.02.007.

Kiniry, J. R., Schmer, M. R., Vogel, K. P., & Mitchell, R. B. (2008). Switchgrass biomass simulation at diverse sites in the northern Great Plains of the U.S. *Bioenergy Res., 1*(3-4), 259-264. http://dx.doi.org/10.1007/s12155-008-9024-8.

Kurka, T., Jefferies, C., & Blackwood, D. (2012). GIS-based location suitability of decentralized, medium-scale bioenergy developments to estimate transport CO_2 emissions and costs. *Biomass Bioenergy*, *46,* 366-379. http://dx.doi.org/10.1016/j.biombioe.2012.08.004.

Laird, D. A. (2008). The charcoal vision: A win-win-win scenario for simultaneously producing bioenergy, permanently sequestering carbon, while improving soil and water quality. *Agron. J., 100*(1), 178-181. http://dx.doi.org/10.2134/agrojnl2007.0161.

Lehmann, J., Gaunt, J., & Rondon, M. (2006). Bio-char sequestration in terrestrial ecosystems: A review. *Mitigation and Adaptation Strategies for Global Change, 11*(2), 403-427. http://dx.doi.org/10.1007/s11027-005-9006-5.

Luman, D., & Tweddale, T. (2008). Assessment and potential of the 2007 USDA-NASS cropland data layer for statewide annual land cover applications. INHS Technical Report 2008 (49). Champaign, Ill.: Illinois Natural History Survey.

Ma, J. G., Scott, N. R., DeGloria, S. D., & Lembo, A. J. (2005). Siting analysis of farm-based centralized anaerobic digester systems for distributed generation using GIS. *Biomass Bioenergy, 28*(6), 591-600. http://dx.doi.org/10.1016/j.biombioe.2004.12.003.

Mohan, D., Pittman, C., & Steele, P. (2006). Pyrolysis of wood/biomass for bio-oil: A critical review. *Energy Fuels, 20*(3), 848-889. http://dx.doi.org/10.1021/ef0502397.

Mullen, C. A., Boateng, A. A., Goldberg, N. M., Lima, I. M., Laird, D. A., & Hicks, K. B. (2010). Bio-oil and bio-char production from corn cobs and stover by fast pyrolysis. *Biomass Bioenergy,*

34(1), 67-74. http://dx.doi.org/ 10.1016/j.biombioe.2009.09.012.
Palma, M. A., Richardson, J. W., Roberson, B. E., Ribera, L. A., Outlaw, J., & Munster, C. (2011). Economic feasibility of a mobile fast pyrolysis system for sustainable bio-crude oil production. *Intl. Food and Agribusiness Mgmt. Rev., 14*(3), 1-16.
Perpiñá, C., Alfonso, D., Pérez-Navarro, A., Peñalvo, E., Vargas, C., & Cárdenas, R. (2009). Methodology based on geographic information systems for biomass logistics and transport optimisation. *Renewable Energy, 34*(3), 555-565. http://dx.doi.org/10.1016/j.renene.2008.05.047.
Pordesimo, L., Edens, W., & Sokhansanj, S. (2004). Distribution of aboveground biomass in corn stover. *Biomass Bioenergy, 26*(4), 337-343. http://dx.doi.org/10.1016/S0961-9534(03)00124-7.
Propheter, J. L., Staggenborg, S. A., Wu, X., & Wang, D. (2010). Performance of annual and perennial biofuel crops: Yield during the first two years. *Agron. J., 102*(2), 806-814. http://dx.doi.org/ 10.2134/agronj2009.0301.
Roberts, K. G., Gloy, B. A., Joseph, S., Scott, N. R., & Lehmann, J. (2010). Life cycle assessment of biochar systems: Estimating the energetic, economic, and climate change potential. *Environ. Sci. Tech., 44*(2), 827-833. http://dx.doi.org/10.1021/es902266r.
Rørstad, P. K., Trømborg, E., Bergseng, E., & Solberg, B. (2010). Combining GIS and forest modelling in estimating regional supply of harvest residues in Norway. *Silva Fennica, 44*(3), 435-451.
Schreurs, E., Voets, T., & Thewys, T. (2011). GIS-based assessment of the biomass potential from phytoremediation of contaminated agricultural land in the Campine region in Belgium. *Biomass Bioenergy, 35*(10), 4469-4480. http://dx.doi.org/10.1016/ j.biombioe.2011.09.005.
Shi, X., Elmore, A., Li, X., Gorence, N. J., Jin, H. M., Zhang, X. H., & Wang, F. (2008). Using spatial information technologies to select sites for biomass power plants: A case study in Guangdong Province, China. *Biomass Bioenergy, 32*(1), 35-43.
Simpson, S., Hamann, M., Parnell, C., Emsoff, S., Capareda, S., & Shaw, B. (2007). Engineering of seed cotton transport alternatives. In *Proc. 2007 Beltwide Cotton Conf.* (pp. 1567-1575).
Singh, J., Panesar, B. S., & Sharma, S. K. (2008). Energy potential through agricultural biomass using geographical information system: A case study of Punjab. *Biomass & Bioenergy, 32*(4), 301-307.
Stephen, J. D., Sokhansanj, S., Bi, X., Sowlati, T., Kloeck, T., Townley-Smith, L., & Stumborg, M. A. (2010). Analysis of biomass feedstock availability and variability for the Peace River region of Alberta, Canada. *Biosystems Eng., 105*(1), 103-111. http://dx.doi.org/10.1016/j.biosystemseng. 2009.09.019.
Tittmann, P. W., Parker, N. C., Hart, Q. J., & Jenkins, B. M. (2010). A spatially explicit techno-economic model of bioenergy and biofuels production in California. *J. Transport Geography, 18*(6), 715-728. http://dx.doi.org/10.1016/j.jtrangeo.2010. 06.005.
U.S. Census Bureau. (2009a). Topologically integrated geographic encoding and referencing system. Washington, D.C.: U.S. Census Bureau. Retrieved from ftp://ftp2.census.gov/geo/tiger/.
U.S. Census Bureau. (2009b). Topologically integrated geographic encoding and referencing system: A shape file with county-level roads and streets. Washington, D.C.: U.S. Census Bureau. Retrieved from ftp://ftp2.census.gov/ geo/tiger/TIGER2007FE.
USDA. (2010a). Cropland data layer. Washington, D.C.: USDA-NASS. Retrieved from www.nass.usda.gov/ research/Cropland/ SARS1a.htm.
USDA. (2010b). Quick STATS. Washington, D.C.: USDA-NASS. Retrieved from www.nass.usda.gov/Quick_Stats/.
USDA. (2011). Conservation reserve program. Washington, D.C.: USDA-NASS. Retrieved from http://www.fsa.usda. gov/FSA/webapp?area=home&subject=copr&topic=crp.
Zhang, F. L., Johnson, D. M., & Sutherland, J. W. (2011). A GIS-based method for identifying the optimal location for a facility to convert forest biomass to biofuel. *Biomass Bioenergy, 35*(9), 3951-3961.